<table>
<tr><td colspan="3" align="center">Type: DSC1-4 (Major)
B.Sc(ECS)-II (Semester III)
Course Title: Database Management System
(Paper Code:)</td></tr>
<tr><td colspan="3">Credits: Theory – (2)   Practical’s – (1)</td></tr>
<tr><td colspan="3">Total Lectures: 30 Hrs.   Contact Hrs. (L) : 2</td></tr>
<tr><td colspan="3">University Evaluation: 30 Marks   Internal Evaluation: 20 Marks</td></tr>
<tr><td colspan="3">Course Outcomes:</td></tr>
<tr><td>1.</td><td colspan="2">Gain knowledge of database systems and database management systems software.</td></tr>
<tr><td>2.</td><td colspan="2">Ability to model data in applications using conceptual modeling tools such as ER Diagrams and design database schema based on the model.</td></tr>
<tr><td>3.</td><td colspan="2">Formulate, using SQL, solutions to a broad range of query and data update problems.</td></tr>
<tr><td>4.</td><td colspan="2">Demonstrate an understanding of normalization theory and apply such knowledge to the normalization of a database.</td></tr>
<tr><td>5.</td><td colspan="2">Be acquainted with the basics of transaction processing and concurrency control.</td></tr>
<tr><td>6.</td><td colspan="2">Familiarity with database storage structures and access techniques.</td></tr>
<tr><td>7.</td><td colspan="2">Analyse strengths and weaknesses of the applications of database technologies to various subject areas.</td></tr>
<tr><td>Unit</td><td align="center">Content</td><td>Lect.</td></tr>
<tr><td>I</td><td>Introduction to database management system:
Definition, limitations of the traditional file system, advantages of DBMS, components of DBMS, database users, database structure database architecture- 2-tier and 3-level (schema) tier architecture, instances, and schema, database languages, data independence, types of data models(hierarchical, network, relational, hybrid).
Conceptual design: ER-model: entities, attributes and its types, relationship, relationship set, generalization, specialization, aggregation, Types of Dependencies, Normalization(1NF, 2NF, 3NF, BCNF, 4NF, 5NF), introduction and features of RDBMS, the difference between DBMS and RDBMS, 12 Codd’s rules.
Relational algebra operations: select, project, Cartesian product, union, set difference.
DDL commands: create, alter, rename, truncate, drop.
DML Commands-insert, update, delete, select statements using where clause.
DCL Commands- grant, revoke, user creation: creating users granting and revoking permissions on database objects, rollback, commit and savepoint datatypes</td><td>15</td></tr>
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	Operators: comparison, conditional, arithmetic, logical, set and special operators – in, not in, between, not between, like, not like, is null, is not null. **Built-in functions:** arithmetic, string, date and time, conversion, aggregate and general. **Clauses:** order by, group by, having clause. **Integrity constraints:** importance of data integrity, not null, unique, foreign key constraint, on delete cascade, check, default constraints.	
II	**Sub queries:** purpose and usage of a subquery, type of subqueries- single row, multiple rows, multiple columns, applying group functions in subqueries, in, any, some, all operators in sub queries. correlated sub-queries, handling data retrieval with exists and not exists operators. **Joins:** inner join, outer joins, Cartesian, self-join and lossless join. **Sequence:** creating, retrieving data, modifying, dropping sequences, synonyms. **Index-** Definition of index, advantages of indexing, types of index, creating index, retrieving data using index. **Pseudo columns:** Types of pseudo columns, currval and nextval, level, rowid, rownum. **Views:** types of views, relational views, object views, using views for DML operations, putting check constraints upon views, creation of read-only views, materialized views.	**15**

Reference Books:	
1.	Database System Concepts by Korth Silberschetz
2.	Fundamentals of Database Systems by Elmsari, Navathe
3.	SQL, PL/SQL – The Programming Language of Oracle by Ivan Bayross
4.	Database Management System by Seema Kedar